

CO-DESIGN RESEARCH AGENDA

Four ways to make the retention budget an explicit cross-layer resource

1. Retention-aware eviction

suspect state → **pin / protect** → **preserve evidence**

Trade bounded memory for replayable diagnostic evidence.

2. Failure-aware scheduling

attribution signal → **scheduler** → **deprioritize risk**

Feed reliability evidence back into placement and execution policy.

3. Budgeted observability

SLO / budget → **trace fidelity** → **dynamic capture**

Negotiate diagnostic fidelity against serving cost at runtime.

4. Infrastructure-triggered diagnostics

latency / memory → **early warning** → **start diagnosis**

Use serving anomalies as triggers before task-level failure surfaces.

All four are research proposals grounded in surveyed components; none is claimed as an implemented system.